



WEARABLE SYSTEM FOR MONITORING POSTURE AND HUMAN MOVEMENTS

MS6 2/06/2026 - Group 2



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PROBLEM



WHAT IS THE PROBLEM?

Muscular injuries caused by **poor posture** when lifting weights in daily life.



WHY ISN'T IT SOLVED?

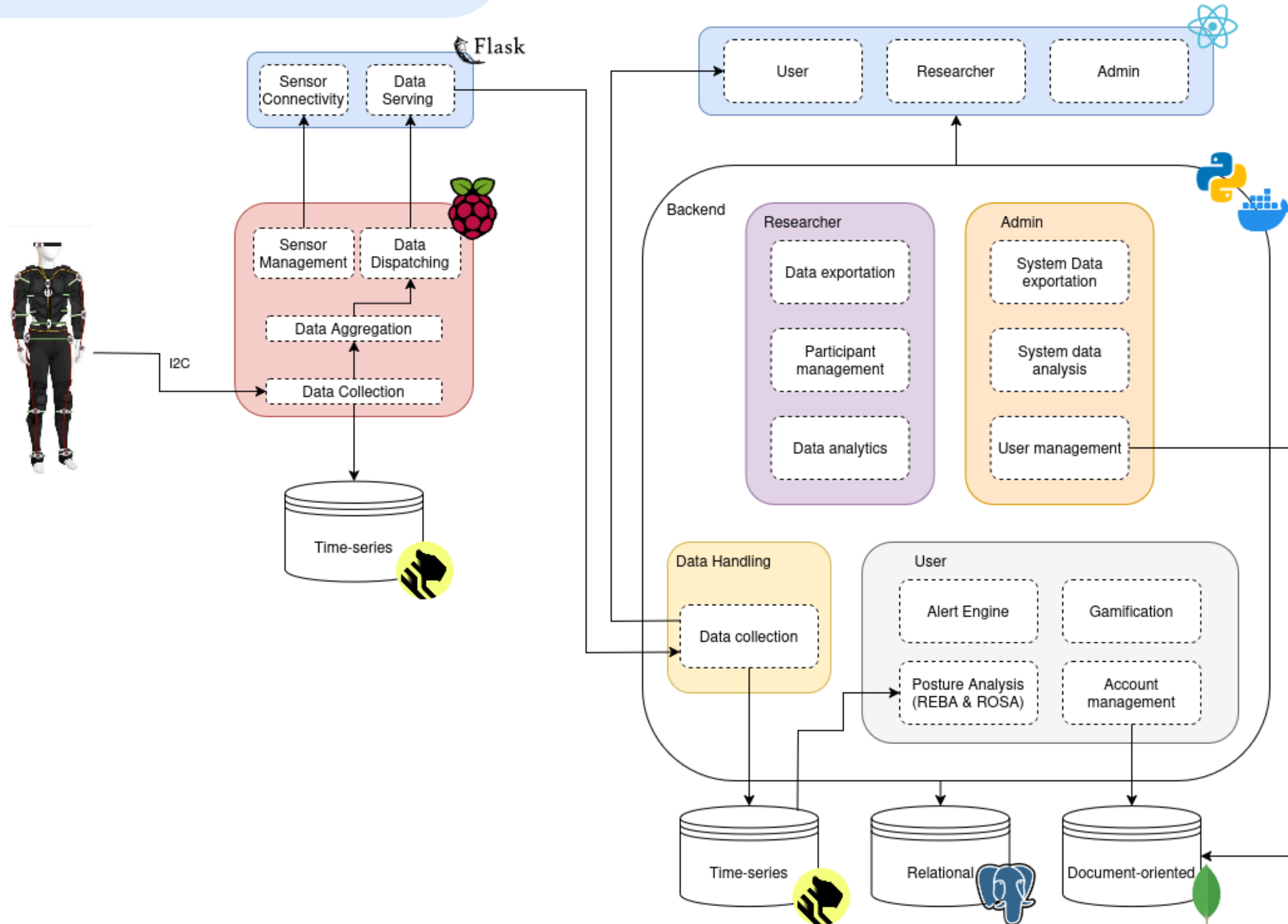
People **maintain** incorrect postures out of habit and for immediate comfort, without being truly aware that they are **making a mistake** or **how to correct** their position.

Motion Suit

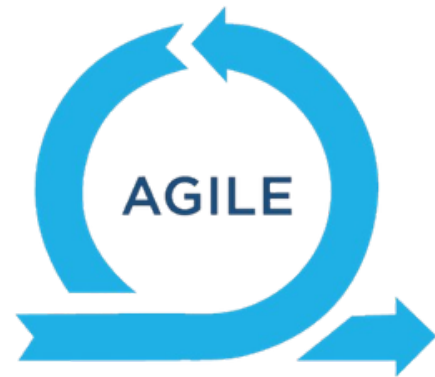
Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies



ARCHITECTURE



METHODOLOGY



Agile Planning



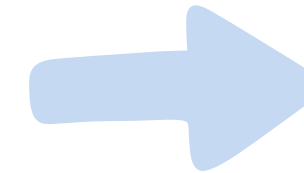
Milestone-Driven
Sprints



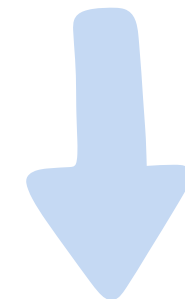
Hardware-dependent
tasks delayed until
sensors arrived

Workflow:

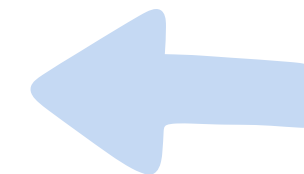
New feature



Opening a
new branch



Branch PR



Push to dev

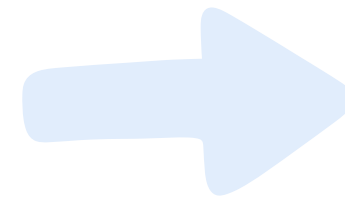


PR to main
(before MS)

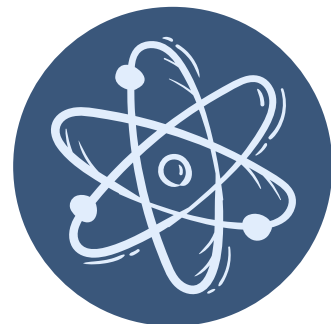
CHALLENGES



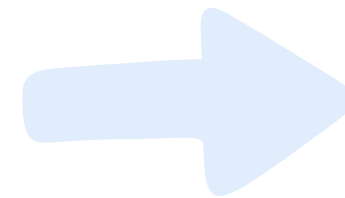
Real-time **communication** between the Raspberry Pi and the backend



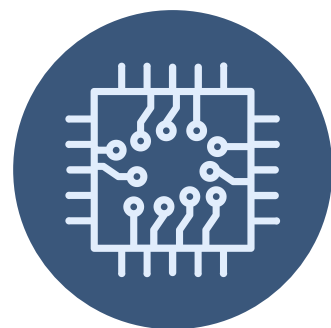
- Increase I2C bus speed on the raspberry Pi
- Improved efficiency of payload injection on the backend



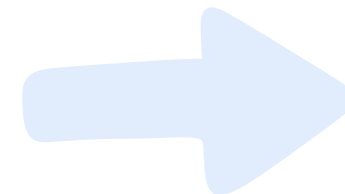
Working with **physics** in data collection



Suit calibration with default quaternions

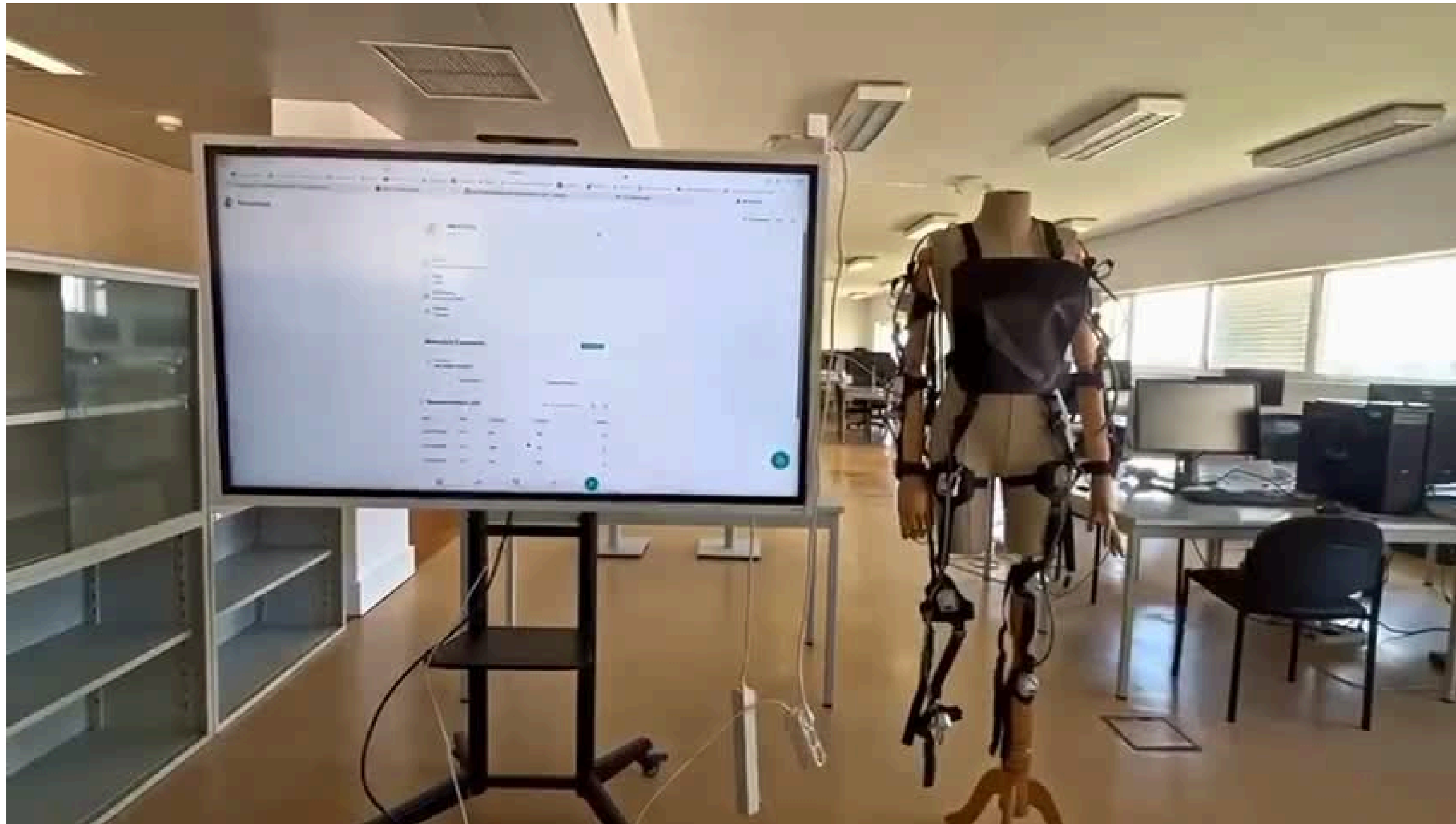


Working with **hardware**



Self research

DEMO



RESULTS

User story

Status

US-1: Real-time incorrect posture alerts



US-2: Posture history and evolution charts



US-3: Real-time metrics and 3D body model visualization



US-4: Customization of alert types (sound/vibration/visual)



US-5: Setting daily and weekly posture goals



US-6: Visual progress tracking and goal streaks



RESULTS

User story	Status
US-7: Simple suit calibration via the mobile app	
US-8: Gamification system with badges and points	
US-9: Full access to documentation, repositories, and system logs	
US-10: User account management and access control	
US-11: Analysis and data export in CSV/JSON formats	
US-12: Data comparison by demographic and professional groups	

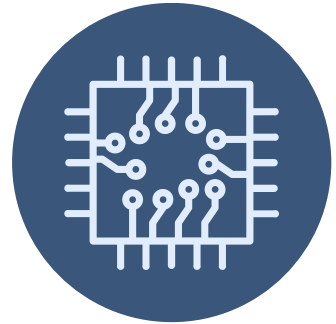
RESULTS

Functional Prototype

Extras	Status
Chatbot JARVIS	✓
Text-to-Speech	✓
Daltonic Mode	✓



FUTURE WORK



Use additional sensing modalities to estimate translational motion and ground-relative position.

Edge-to-Cloud Roadmap

Edge Layer

Raspberry Pi processes real-time posture alerts locally (100% offline)

Cloud Layer

Managed databases (AWS RDS / Timestream) to offload physical host storage

Async Ingestion

Event-driven Message Broker (Kafka / RabbitMQ) to buffer heavy sensor stream

CALENDAR



June 2

MS6: project presentation

T-16

Done

Cable and Sensor Installation



<https://pei-motionsuit.github.io/microsite/>

THANK YOU!



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